Henry Speiser

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Education

University of California, Berkeley — Expected Graduation May 2028
B.S. Electrical Engineering and Computer Science | College of Engineering — *GPA 3.75/4.00*

Experience

Aleph Surgical — San Francisco, CA
Member of Technical Staff Intern — *September 2026 - Present*
- Working on autonomous suturing, surgical teleoperation, and robot learning.

Origami Robotics — San Francisco, CA
Robotics Engineer Intern — *May 2026 – August 2026*
- Started on hand retargeting and traced poor performance back to the hand's kinematics rather than the algorithm. Took over the Rev. 1.5 21-DoF redesign, changing joint axes, finger geometry, and actuator layouts to better reproduce human grasps.
- Existing hand benchmarks were easy to game, so I built grasp-based ones in MuJoCo, sampling millions of object/hand poses to test whether designs could actually grasp and hold objects, then using the results to search a 30-parameter design space.
- Designed, manufactured, and assembled the next-gen hand in CNC'd aluminum and 316L stainless, including actuator sizing and 48V three-phase axial-flux motor controllers with magnetic encoders and CAN small enough to fit inside the fingers.
- Built the synchronized teleoperation/data-collection stack around Manus gloves, Vive trackers, Raspberry Pi cameras, and robot joint states; also worked on fingertip tactile sensing.
- Worked on S1, Origami's data collection glove, whose 20 encoded finger joints map 1:1 onto the robot hand so demonstrations need no retargeting; it also records fingertip/palm pressure and video in sync. Early units were sold to OpenAI.

Tesla — Palo Alto, CA
Low Voltage Electrical Engineer Intern — *September 2025 – December 2025*
- Worked on Cybercab low-voltage systems including the windshield controller, TPMS, displays, Bluetooth, and GNSS, building fixtures and root-causing failures through thermal and vibration testing, including camera fogging in a room-sized thermal chamber.
- Every test had its own automation stack, so I built a modular framework where engineers could plug reusable hardware-control modules into one orchestrator. It cut lifecycle stress-testing time by 25% and was adopted across the org.
- Built Python pipelines that aggregated thousands of hours of electrical logs in real time and pushed results directly into Jira, making failures visible while tests were still running.

Somo AI — San Francisco, CA / Mexico City, MX
Hardware & Firmware Engineer Intern — *April 2025 – September 2025*
- Joined when Somo was about five people and had no hardware prototype, so I designed the first one from scratch. That board became the platform for figuring out the product's electronics, packaging, sizing, and field requirements.
- From there I designed 12+ boards across ESP32, STM32, Nordic, and Raspberry Pi, often shipping one or two revisions every two weeks; integrated mmWave, thermal, IMU, PIR, Ethernet, current sensing, CAN/RS-485, BLE, and 915 MHz radios.
- Wrote the RTOS firmware, tested the finished products, and designed enclosures; thousands of the resulting boards are now deployed in hotels around the world, including 1,000+ production units of my PIR sensor.

Atila Biosystems — Sunnyvale, CA
Electrical & Firmware Engineer Intern — *October 2022 – December 2023*
- Came in with one goal, to recreate a $600 PCR-testing device for roughly $25 so it could go into homes and low-resource settings. Designed the PCB, power system, thermal hardware, and rugged solar/battery/USB-C enclosure around low-cost parts.
- Wrote embedded C for real-time PID thermal cycling; designed the heater electronics and state machines around the narrow temperature window needed to amplify biological samples without overheating them.

Projects & Leadership

FRC Team 971 (Spartan Robotics) — Mountain View, CA
Electrical & Software Captain — *August 2021 – June 2024*
- Hard impacts kept killing connectors and sensor wiring, so I made it my mission to eliminate the problem. I designed 11 PCBs for encoders, Hall sensors, beam breaks, and ToF, plus power, CAN, and vision interfaces for early-access Jetson Orins.
- Also wrote autonomous, superstructure, collision-avoidance, and auto-aim software with camera/IMU/encoder fusion; our 2022 autonomous ranked 2nd worldwide out of roughly 4,000 teams.

Supersonic Rocket — Los Altos, CA
Lead Engineer — *June 2024 – September 2024*
- A friend and I wanted to learn rocket propulsion from first principles, so we built nearly the entire vehicle ourselves, including APCP motor grains, thrust stands, a machined converging-diverging nozzle, avionics, wireless ignition, nose cone, and fins.
- Static-fired and modeled the motor before flying to Mach 1.2 and 3,250 ft, with measured motor and flight performance closely matching our simulations.

Skills & Interests

EDA KiCad, Altium, LTspice, ANSYS
CAD & Sim Onshape, SolidWorks, CATIA, MuJoCo/URDF
Languages Python, C/C++, Assembly, Java
Protocols CAN/RS-485, SPI/I2C, Sub-GHz (LoRa/OOK), BLE
Firmware RTOS, Embedded Linux, Sensor Fusion, PID, FOC
Interests Ski Racing, Water Polo, Backpacking, Climbing, Movies